

Recent advancements using machine learning & deep learning approaches for diabetes detection: a systematic review

Neha Katiyar^{*}, Hardeo Kumar Thakur, Anindya Ghatak

Bennett University, Greater Noida, India

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ABSTRACT

Nowadays, Diabetes Mellitus is one of the significant health challenges that affects many people across the world. Early detection of Diabetes Mellitus will help in preventing complications, i.e., kidney disease, nerve damage, eye damage, etc. Over the past few years, several Machine Learning and Deep Learning techniques have been applied for the early detection of Diabetes Mellitus. The paper provides reviews on various Machine Learning and Deep Learning techniques applied for early detection of Diabetes mellitus. The review criteria mainly focus on five topics: the diabetes dataset, methods used, performance metrics, limitations of the work, and the overall status of diabetic research. The objective of this paper is to provide a comprehensive review of Diabetes Mellitus prediction techniques applying Machine Learning and Deep Learning that will be helpful sources for researchers in the healthcare field.

1. Introduction

In the realm of healthcare, it is crucial to anticipate illnesses and facilitate their prompt identification. The anticipation of diseases becomes feasible with a comprehensive understanding of their symptomatic manifestations. Such diseases can stem from genetic predispositions or can be attributed to individuals living conditions. Presently, there is a growing inclination towards adopting opulent lifestyles, which can disrupt the body's hormonal balance, leading to a myriad of hormonal irregularities, including ailments like diabetes. A considerable number of individuals are grappling with such ailments at an early age, contributing to a substantial repository of electronic health data. Employing diverse data mining methodologies and analyses, the healthcare system strives to harness this extensive information. This comprehensive information exerts a meaningful influence on effectively regulating and handling the issues of diabetic persons. Al Jilailaty et al. [7] performed diabetes detection was quickly with the dataset. The dataset set consists of specific parameters that are identified soon, like glucose level, Body mass Index, etc. These datasets are refined with data mining techniques. The domain of data mining in healthcare constitutes an expansive field of study encompassing a wide range of machine-learning techniques, such as Support Vector Machines (SVM), Regression, k-nearest Neighbours (KNN), and Artificial Neural Networks (ANN), among others—consequently, the progression in machine

learning methodologies in early-stage detection and diagnosis of diabetes through automated systems.

Diabetes datasets play a crucial role in early-stage detection. Ahmed et al. [1] have mentioned in the article that patient record is obtained from electronic devices and paper records, with the automotive record having timestamps event and the paper record having only "logical time" slots. The diabetes datasets help to detect diabetes in a person. Attributes of a dataset play a crucial role in detection. Researchers and scientists use different data files to predict diabetes. The ML-based model is required for relevant datasets having essential aspects for training and validations. Choosing the appropriate elements from the dataset increased the ability of the ML model to predict accurate data. The machine learning model is also deliberate in terms of efficiency and effectiveness and is attributed to the transparency of a user. In recent days, many research papers have been published on diabetes detection, interpretation, prevention, and glucose management with machine learning.

1.1. Brief of diabetes mellitus

Diabetes Mellitus occurs in humans and animals. Diabetes affects 1.5 million people in a year. Thakkar et al. [10] have provided the info about 48% of people in this world die due to diabetes mellitus. This disease occurs when the disorder of carbohydrate metabolism shows the character of impaired ability in the human body not producing enough

^{*} Corresponding author.

E-mail address: e22soep0071@bennett.edu.in (N. Katiyar).

Nomenclature	
DT	Decision Tree
SVM	Support Vector Machine
DL	Deep Learning
ML	Machine Learning
PCA	Principle Component Analysis
DM	Diabetes Mellitus
ANN	Artificial Neural Network
KNN	K-Nearest Neighbour
NB	Naive Bayes
PIDD	Pima Indian Diabetes Dataset
RF	Random Forest
CNN	Convolutional Neural Network
ETL	Extraction, Transformation and Load
ECG	Electrocardiogram Signals
RNA	Ribonucleic acid
TD1	Type-1 Diabetes
TD2	Type-2 Diabetes
LSTM	Long short-term memory
IoT	Internet of Things
RNN	Recurrent Neural Networks

quantity of insulin. The supply of insulin level is not maintained in the human body, as mentioned in the article Siddiqui et al. [9].

Diabetes results from a malfunctioning pancreas in the human body, as mentioned in the article Alkhodari et al. [6]. For example, when the human pancreas is not producing insulin as it should. The human body does not supply insulin adequately. A person may get affected by diabetes due to three main reasons: genetics, surroundings, and an affluent lifestyle. Lifestyle is an essential factor in becoming a person diabetic and non-diabetic. If a person's ancestor is non-diabetic previously but the person has diabetes, then this has to be done by an affluent lifestyle. The next cause is the environmental impact. Now, people are influenced by the effects of losing weight. To lose weight people are taken poor diet and frequent exercises. That causes the problem of renal failure. Renal failure is the first step towards post-diabetes. In this modern world, people are facing issues like frequent hunger, frequent urination, weight loss and vision loss due to these signs of diabetes.

The benevolence of this review paper in the domain of diabetic research considered ML-based access in DM detection, prevention, self-management, and personification. A review paper plays a crucial role in the study because it efficiently summarizes cutting-edge research in a specific area. The demographic information from ML approaches diabetes-related health measurements and the risk factors easily calculated models. The ML model is used for long-term accuracy calculations given by Lekha et al. [5]. It is the main objective of ML models. The objectives of the Diabetes Mellitus Machine learning model of this manuscript are given as follows-

1. The repository's publicly available datasets are used to explore the different types of datasets used in DM detection.
2. The pre-processing method applies to Diabetes Detection.
3. Machine Learning-driven performance in detection and classification tasks.
4. Deep learning methods for Classification, Detection, and Diabetes Management.
5. Performance metrics for diabetes detection and diagnosis algorithms.
6. The limitations in the ML model for diabetes detection and prevention.

This manuscript is classified as follows- Section 2 has a precise

description of Diabetes and the types of Diabetes. Section 3 provides machine Learning and different classification techniques. Section 4 provides a brief about the survey paper criteria. Section 5 shows the systematic review of the dataset pre-owned by researchers. Section 6 provides an overview of ML and Deep Learning approaches in the area of Diabetes Mellitus. This section also consists of an analysis of various performance measures and limitations. Section 7 is based on the various discussion topics. Section 8 has future scope and challenges, and the conclusion is summarized from the paper.

2. Types of diabetes

There are four types of diabetes, as mentioned in Fig. 1-

2.1. Type-1 diabetes

Type-1 diabetes is found in the human-resistant body. It is a chronic autoimmune disease that perturbs the pancreas. In Type-1 diabetes, the immune system attacks and damages insulin, producing beta cells in the pancreas. Insulin regulates blood sugar (glucose) levels by enabling the absorption of blood sugar into cells, where it is utilized for energy production. When the beta cells are abolished, the body is unable to regenerate insulin, leading to elevated blood sugar levels.

Without insulin, the glucose level rises. To prevent this type of diabetes, people have to take insulin every day to maintain their glucose levels, and that keeps them alive. The problems that arise with Type-1 diabetes are heart disease, strokes, dental disease, foot problems, depression, and kidney diseases as mentioned in the article, Zhu et al. [22]. The symptoms of this diabetes, as seen in the human body, are excessive thirst, frequent urination, extreme hunger, unexplained weight loss, fatigue and blurred vision.

2.2. Type-2 diabetes

Type-2 diabetes happens when the human body regulates sugar as fuel. The sugar is called glucose. When sugar circulates in the human body for a long time, it raises the blood sugar level in humans. The high sugar level discards the human immune system, circulatory systems, and nervous systems. In Type-2 diabetes, through the pancreas, good enough quality insulin is available, although the insulin does not work correctly because of the rise in sugar levels as mentioned in the article Kazerouni et al. [15]. Type-2 diabetes causes problems like increased thirst, frequent urination, fatigue, and areas of darkened skin, usually armpits and necks.

2.3. Pregestational diabetes

Pregestational diabetes is the combination of Type-1 and Type-2 diabetes that occurs in females during the time of pregnancy. Pregnant women have pre-existing diabetes during the time of pregnancy. According to Fazakis et al. [8] pregestational increases the risk of malfunctions in the baby during the growth process in the ovary. The result of this type of diabetes is the prematurity of the baby, and operative delivery takes place. The diabetic women proceed to cesarean section. Women with pre-gestational diabetes were overweight and older. Diabetes occurs during this pre-gestational time and lasts longer in women. This diabetes occurs in women after pregnancy.

2.4. Gestational diabetes

Gestational diabetes is detected for the first time during pregnancy. Gestational diabetes affects the glucose level of pregnant women as well as the body. It can also affect the baby's fitness. During the time of pregnancy, gestational diabetes can easily be controlled by eating healthy food, exercising, and taking proper medication. The symptoms of diabetes do not quickly happen, given some unnoticeable signs occur,

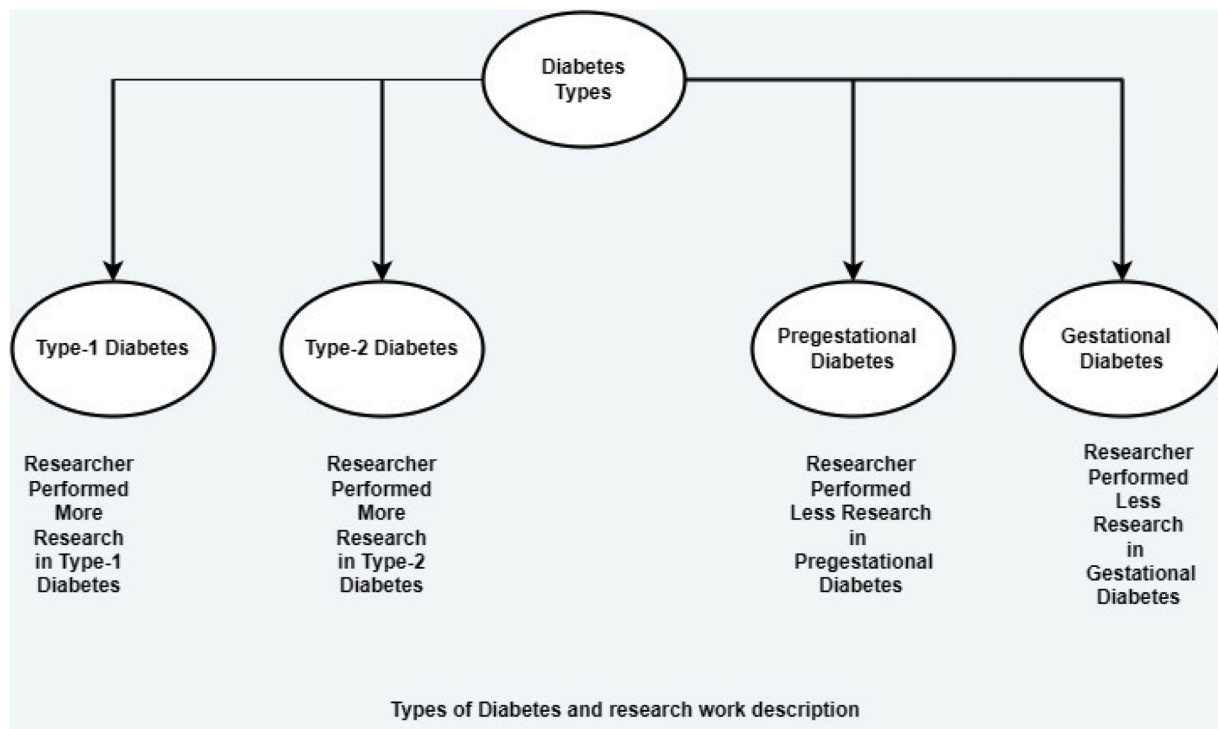


Fig. 1. Types of Diabetes and description of research work.

like increased thirst and more frequent urination studied by Suyanto et al. [12]. In this type of diabetes, a monthly check-up is recommended by the doctor.

3. Types of machine learning techniques

The Machine Learning algorithms are categorized into four sections, as shown in Fig. 2.

- 1) Supervised learning, which operates on labelled data.
- 2) Unsupervised learning, which works with unlabeled data.
- 3) Semi-Supervised learning, which utilizes both labelled and unlabeled data.
- 4) Reinforcement learning, which operates based on feedback systems and experiences as given in article Shokrehkhodaei et al. [2].

3.1. Supervised learning

The supervised learning worked on the labelled datasets. The

labelled data have similar features and characteristics. The machine learning procedure supervises the outputs of a given set of inputs. In this algorithm, suppose that the input variable of the diabetic dataset is X, and the output variable is Y. When we apply this learning algorithm, we get the predicted new input as X. The latest output is Y given by [16].

The semi-supervised algorithms used for diabetes detection are KNN, DT and SVM. It is an effective approach where the model is trained for the labelled data, meaning the input features are associated with the diabetes outcomes. It also uses overfitting techniques. Additionally, the supervised algorithm performs feature selection and extraction techniques to enhance the model interoperability.

3.2. Unsupervised learning

Unsupervised learning uses unclassified, unlabeled data. The main objective of unsupervised learning is to acquire unhidden information from datasets. We can get unhidden information from unique training systems or add the classification process from the datasets. This learning applies clustering techniques and divides the data according to the features from a group of items that have similar features.

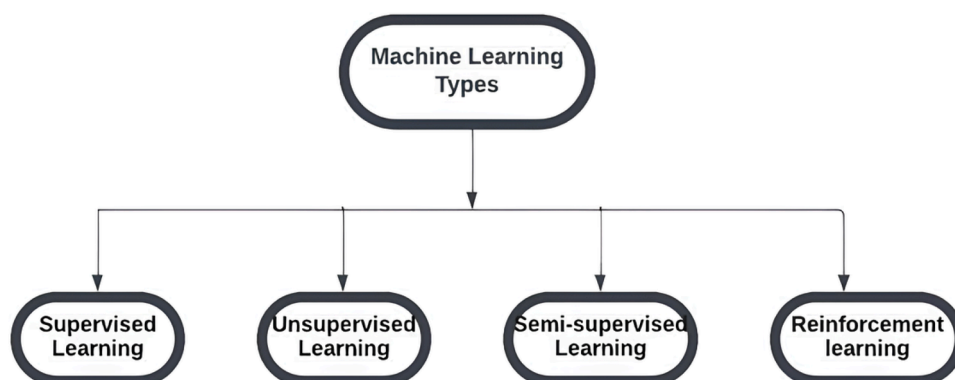


Fig. 2. Types of Machine Learning Techniques.

The unsupervised algorithms used for diabetes detection are PCA, Neural Networks, Hierarchical Clustering, and K-mean Clustering. Lekha et al. [5] performed research using the techniques of diabetes detection use clustering, dimensionality reduction and generative models.

3.3. Semi-Supervised learning

The ML algorithm is an intermediate between supervised and unsupervised learning algorithms [3-4]. It consolidates both labelled and unlabeled data. In this algorithm, the training data is in the form of tuples. Various labels are applied to the dataset to easily provide the predicted output. If we apply a pseudo-labeling process, the output may be incorrect.

The Supervised algorithm used for diabetes detection is clustering, densities and dimensionality reduction. In semi-supervised learning, diabetes detection is performed with fine-tuning, Interference, Monitoring and Updating.

3.4. Reinforcement learning

Reinforcement learning uses feedback systems for learning. In this learning procedure, the Agent learns and improves its performance by using feedback systems. It does not use any labelled data. It solves the problem of sequential decision-making. In this learning, the output and

label are not pre-defined. It totally worked on the hit-and-trial procedure. The Agent's action worked according to the previous action. Ghosh et al. [18] focused on the reinforcement learning procedure mainly depends upon the Agent, environment, action, reward and policy.

The reinforcement learning used for diabetes detection is Thompson sampling and posterior sampling. The main goal of reinforcement learning is to optimize the insulin dosage for a patient with diabetes and minimize the risk of diabetes hypoglycemia (low blood sugar level).

4. A brief about the survey paper criteria

In the healthcare sector, particularly within the realm of diabetes mellitus, many works are available. The work chosen for our review process is sourced from Scopus or may originate from peer-reviewed journals. Their primary purpose lies in aiding academic researchers and scientists in the early-stage prediction of diabetes. Additionally, these research papers may serve as a foundation for the development of advanced techniques for diabetes prediction. Notably, these papers straddle two prominent domains: medical science and computer science. Chaki et al. [17] has given a prevalent theme in diabetes detection research involves the intricate interplay between Machine Learning (ML) and Deep Learning (DL) techniques. These intelligent computational methods represent cutting-edge approaches in the healthcare field, offering a wealth of precise manuscripts that harness the power of

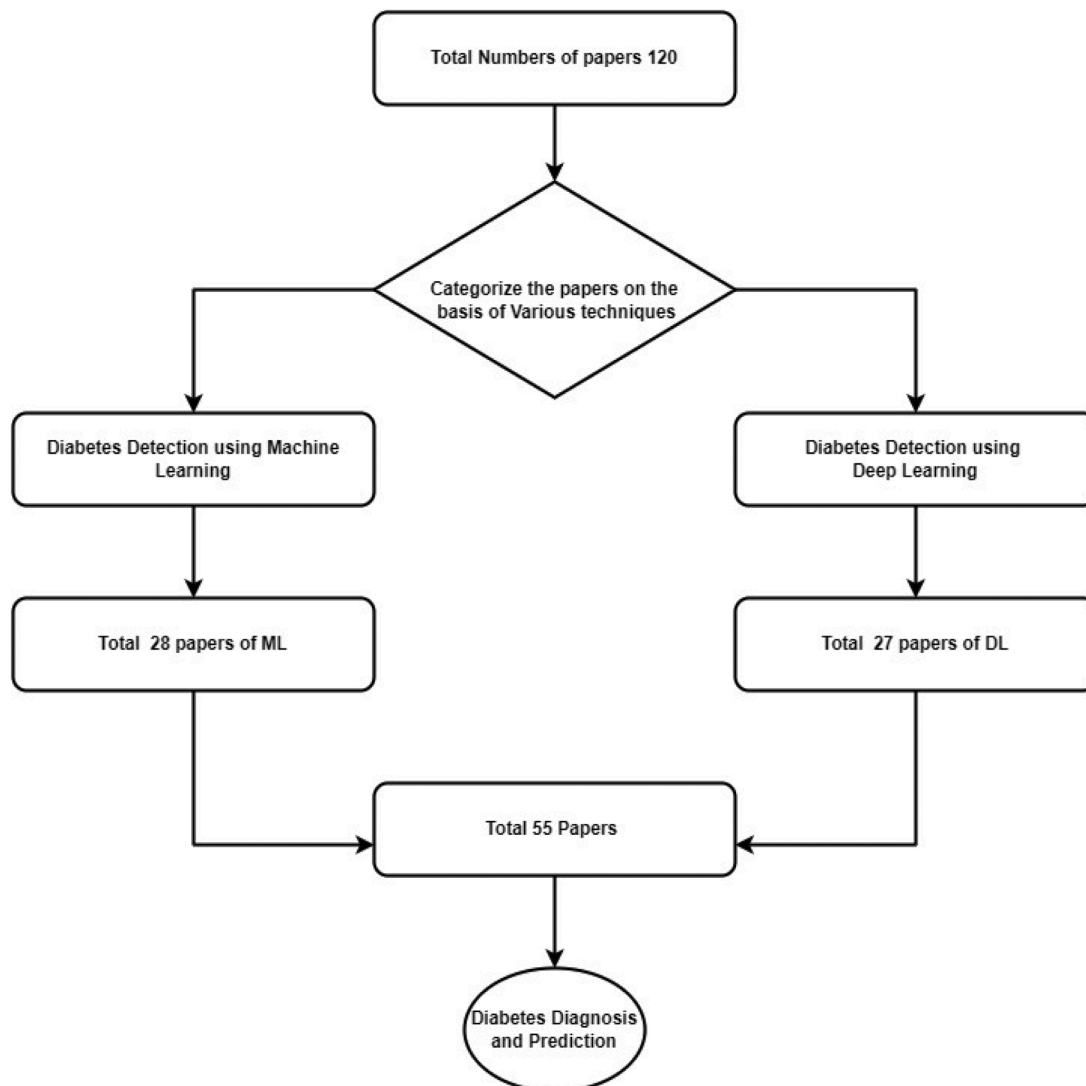


Fig. 3. Paper Selection Process for Review.

these techniques.

Although published a large number of articles based on these two advanced techniques, our scope of research has been restricted over the last seven years (2018 to 2024). It is compulsory to publish many articles. However, the articles are bounded to the areas of ML and DL, although the wider area of computer science.

The adjacent phase performed a systematic review of ML techniques and retrieved all the articles related to the area. It also removed those papers that did not consist of the field of diabetes mellitus using Machine Learning. This paper cumulated the review of 55 papers. The review paper was categorized into classification, detection, and diagnosis of disease. The absolute collection of papers is shown in Fig. 3.

5. Datasets used by researchers

Datasets play an essential role in finding the results in machine learning. A dataset is a collection of data that has similar values and features. In the context of tabular data, which has similar features, tabular data is represented in the form of rows and columns. A column represents a particular variable. When it comes to diabetic datasets, many sampling techniques are applied, and a large number of observations are performed.

To perform diabetic research on a dataset. The data must be normalized, and the data must be divided into pre-processing states. In this study, we have gone through various types of datasets. The Pima Indian diabetes dataset [PIDD] is the most popular dataset used by researchers in the study of diabetes detection and prediction. Tripathi et al. [19] mentioned in the article the PIDD data contains information on 768 women and their health records. This dataset was given and verified by the National Institute of Diabetes and Digestive and Kidney Diseases. The dataset was taken from women who were at least 21 years old. Diabetes datasets are available in many forms, including numerical data, alphanumerical data, and image datasets, as shown in Table 1. Each type of dataset uses so many adaptive techniques like segmenting, thresholding, sampling bagging and boosting. In some diabetic-based research, data collection may be done through an online survey option. In some research, the data may be used as filtered data, and other researchers may use unfiltered data. Some researchers may remove the missing values from the data, and some use the original data and focus on machine-learning techniques. Nnamoko et al. [41] have performed the class imbalance procedure. To combat this, they implemented SMOTE algorithms. Additionally, they apply data pre-processing techniques to enhance the performance of training data by artificial minority instances within the training data. It performs classification and bias towards standard cases. Kumar et al. [42] explored class-imbalanced learning scenarios. They meticulously examined a total of 51 articles related to imbalanced data. Within their work, they explored seven distinct class-balancing techniques.

Diabetes detection on imbalanced data. Wang et al. [43] opted the naive Bayes method to normalize data with missing values. Furthermore, they harnessed the adaptive synthetic sampling method to mitigate class imbalance issues. To validate their methodology, they applied these techniques to the Pima Indian Diabetic Dataset. The significance of machine learning and deep learning approaches in developing diabetes detection systems, the need for improved data availability, the potential of reconstructing detection models based on non-invasive measurements, and the importance of selecting appropriate datasets for training models to address the rising prevalence of diabetes worldwide [52]. Al-Absi et al. [53]. worked on the enhancement of diabetes diagnosis accuracy by developing an improved predictive model using retinal images from the Qatari population, achieving an accuracy of over 92% with the DiaNet v2 model.

Diabetes datasets are the base of research in the field of prediction and detection of diabetes. There can be so many types of data collection, such as open access, survey-based data collection, and self-created data collection, as mentioned in the article by Ye et al. [20]. Data can be

Table 1
Diabetes dataset description with references.

S. No.	Datasets	DatasetDescription with links	References/ Published Year
1	Pima Indian Diabetes Dataset (PIDD)	The Pima Indian Diabetes Dataset consists of data from 768 women. This data is freely available on the UCI repository and Kaggle. The dataset encompasses variables such as blood glucose levels, insulin, body mass index, and more. It consists of eight independent variables, with one being categorical and the remaining seven continuous. Therefore, at a minimum, eight plots are required. To extract meaningful insights and ensure a thorough analysis, the target variable result is plotted against each independent variable if we want to derive any inferences and leave no stone unturned for it.	Ahmed et al. [1], Sharma et al. [3], Thakkar et al. [10], Suyanto et al. [12], Mamatha et al. [13] Khanam et al. [16], Gosh et al. [18], Tripathi et al. [19] Ayon et al. [27], Aada et al. [29], Pranto et al. [30], Olisah et al. [37], Yahayoni et al. [39]
2	Signals Data Wavelength Data	The dataset consists of the wavelength of blood tissue level between various nm. The whole wavelength measures the intensity between 410 and 990 nm. The intensity of four wavelengths is (485, 645, 860 nm, and 940 nm).	Shokrehodaiei et al. [2]
3	Medical Information Mart for Intensive Care (MIMIC-IIIrd)	This dataset is publicly accessible. It is a relational database that contains the data of unidentified patients. The database includes information on 38,597 adult patients who were admitted between 2001 and 2012.	Lekha et al. [5]
4	Bangladesh Institute of Health and Sciences (Diabetic Datasets)	This dataset includes 95 patients in the Bangladesh Institute of Health and Sciences from December 18 to 2017 and April 26 to 2018. There are 45 Males and 50 females in this dataset.	Alkhodari et al. [6]
5	OREBA FIC CLEMSON	The OREBA stands for (Objectively Recognizing Eating Behavior and Associated Intake). The objective of the OREBA dataset is to consist of a multi-sensor recording of communal intake occasions for researchers hooked on the automated exposure of absorption gestures. The gestures study is easily performed with this dataset. The Multimedia Understanding Group provided the Food Intake Cycle (FIC) dataset to determine in-meal eatingbehaviour. The CLEMSON consists of data from 271 participants with 488 recordings. Each	Al Jlailaty et al. [7]

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Table 1 (continued)

S. No.	Datasets	DatasetDescription with links	References/ Published Year
6	FINDRISC ELSA	consumption gesture, including hand, utensil, container, and food, is easily collected and monitored with pre-processing. FINDRISC is a diabetic risk assessment tool that is an effective undiagnosed tool for Type-2 Diabetes Mellitus. The ELSA stands for (English Longitudinal Study of Aging). The ELSA database is a standard that can stratify train and test split procedures. It can preserve the dataset with fixed class properties.	Fazakis, et al. [8]
7	PubMed Survey Database taken from the year (2012–2016).	PubMed is a Digital Library and Springer for non-invasive blood glucose monitoring from 2012–2016. After finding the datasets for the keywords non-invasive blood glucose self-monitoring and invasive monitoring, the research data can be used in biomedical, computer science, and related fields.	Siddiqui et al. [9]
8	Sri Pamela Hospital Kumpulan Plane Hospital Indonesia	The dataset has 158 records and 15 attributes. The ETL process is performed on the database to obtain the eight attributes required for patient information and diabetes detection.	Fiarmi et al. [11]
9	Survey Based database from Kashmir Valley	The Kashmir Valley dataset has a set of 734 patients of all age sets. The dataset collected in the Kashmir Valley was taken from one diagnostic lab. The dataset has 11 attributes: Age, Plasma Glucose Fasting, Post- Prandial Glucose Level, Body Mass Index, Diastolic Blood Pressure, waist thickness, and diabetic history of the patient included other attributes.	Shuja et al. [14]
10	ShohadaHospital Tehran	Diabetes data was collected from 200 unrelated diabetic Iranian patients, 100 T2DM patients, and a comparative study between 100 healthy persons paired for age and sex. This diabetic study was also based on six attributes. T2DM patients were enlisted from individuals belonging to the Diabetic Clinic at Shohada Hospital, Tehran, Iran.	Kazerouni et al. [15]
11	Hainan Medical University China	The Department of Respiratory Disease datasets gathered breath samples for the detection	Ye et al. [20]

Table 1 (continued)

S. No.	Datasets	DatasetDescription with links	References/ Published Year
12	University California Irvine gas sensor datasets	of diabetes. The dataset consists of approximately 210 measurements obtained from 41 participants (21 females and 20 males). Each participant sample was collected each day for the dataset. The dataset consists of 62 sensors at various concentrations. One concentration has a sample size of 20* 7. The training set selected 44 samples for each concentration.	Zhu et al. [21]
13	OhioT1DM ABC4D ARISES	In this article, three datasets are used- 1)The first one is the OhioT1DM dataset, which is publicly available and contains data on Type-1 diabetic patients. 2)The second dataset is the ABC4D dataset, which consists of the data of 25 patients with Type-1 diabetes. The dataset contains information on a six-month clinical period. 3)The third dataset is ARISES. It consists of data from 12 patients with Type-1 diabetes. The data set has information on the 12 weeks of the 12 patients.	Zhu et al. [22]
14	ECG signal data on 218 patients	This research data has 86 diabetic personrecording to be done for the automatic screening algorithm of diabetes mellitus. They select some features after that, apply the Kruskal wall test.	Gupta et al. [23]
15	RNA Sequencing Data	The RNA sequencing dataset has 1600 human pancreatic cells. In this data set, 651 people have non-pancreatic datasets, and 941 people have TD1 diabetic pancreatic cells.	Li et al. [24]
16	Self-Created database research collection of data for Type-1&Type-2 diabetes	It is survey-based research on diabetes. Researchers collected the data and named this project PROMs. The National Health Service (NHS) site in the United Kingdom (UK). Due to COVID-19 constraints, interviews will be taken remotely (either via web chat or telephone) according to persons' preferred availability on date, time, and medium. In this, they collected the data from 1000 participants.	Carlton et al. [25]
17	Self-Createddataset collection	This dataset consists of 276 persons the study is based on the age group between (18 to 75) years. The clinical follow-up was taken of the person, whether the person is	Aguilera et al. [26]

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Table 1 (continued)

S. No.	Datasets	DatasetDescription with links	References/ Published Year
18	National Health and Nutrition Dataset	diabetic or non-diabetic; it covers six months. This dataset consists of 6561 people. In this dataset, 657 are diabetic, and 5904 are non-diabetic. This study consists of 50% of male data. This study used an open data survey based on a study of the US population.	Maniruzzaman et al. [28]
19	Clinical data collection by OpASHA	This clinical data collection procedure took place between 2013 and 2015. They took data from 336 patients with diabetes, 86% of whom were diagnosed with a blood glucose level.	Gröschel et al. [31]
20	National Institute of Diabetes and Digestive and Kidney Diseases	Researchers took the diabetes dataset from the National Institute of Diabetes and Digestive and Kidney Diseases and tested patients for diabetes or non-diabetes. Of this diabetic data, 506 are training samples, and 262 are testing samples, and they took a total of 9 attributes to perform the research.	Yasen et al. [35]
21	Sree Diabetes Diagnostic center Kurnool	The diabetic data was taken from the Sree diabetic center Kurnool in Andhra Pradesh. It was collected from September 2014 to January 2015. The data was collected from the age group 18 to 77. The data has been measured in various attributes. The diabetes status is estimated on each class label.	Thipa Reddy et al. [36],
22	Self-Created Data Collection from Jammu& Kashmir Valley	This research data was collected over two years. The data was collected from 2019 to 2020. The data was collected in the Jammu Kashmir valley from different departments, such as inpatient, outpatient, emergency case, etc. The data collected from both regions like urban and rural.	Ganie et al. [38]
24	Self-Created Small dataset on Skin Thickness	The experimental research data on the skin is from 20 patients. The data has the skin information of 10 men and ten women. The data was taken from (11 to 54 years). The measurement of skin performed on the dorsal surface of the foot.	Dremin et al. [40]

collected by respiratory sensor. Data was collected from various blood tissue levels and their wavelengths. Each piece of data provides information according to the research and its experimental process. There are so many datasets that are used for diabetes detection and prediction. These datasets are available with links, as shown in Table 2. Some frequently and rarely used datasets are found in Table 1, which not only

Table 2

Datasets with associated download links and websites.

Datasets	Websites& Links
Pima Indian Diabetes Dataset (PIDD)	Link-Pima Indians Diabetes Database (kaggle.com)
Signals Data	Link- A dataset of hemoglobin blood value and photoplethysmography signal for machine learning-based non-invasive hemoglobin measurement - ScienceDirect
Wavelength Data	Link- MIMIC-III, a freely accessible critical care database - PMC (nih.gov)
Medical Information Mart for Intensive Care (MIMIC-IIIrd)	Link- Field Research and Practice Centre (FRPC) - Bangladesh University of Health Sciences (buhs.ac.bd)
Bangladesh Institute of Health and Sciences (Diabetic Datasets)	Link- 1)OREBA Dataset Papers With Code
OREBA	2) The Food Intake Cycle (FIC) Dataset Multimedia Understanding Group (auth.gr)
FIC	3) All Data Sets Data Sets Clemson University
CLEMSON	Link-1) FINDRISC Diabetes Risk Calculator QxMD
FINDRISC	2) Accessing ELSA data ELSA (elsa-project.ac.uk)
ELSA	Link- PubMed (nih.gov)
PubMed Survey Database taken from the year (2012–2016).	
Sri Pamela Hospital	Link- Indonesian hospital database - Mendeley Data
Kumpulan Plane Hospital Indonesia	
Survey Based database from Kashmir Valley	Link- Prevalence and Early Prediction of Diabetes Using Machine Learning in North Kashmir: A Case Study of District Bandipora - PMC (nih.gov)
	Link- Prevalence, awareness, treatment and control of diabetes among Iranian population: results of four national cross-sectional STEPwise approach to surveillance surveys - PMC (nih.gov)
Shohada Hospital Tehran	Link- Hainan Medical University (hainmc.edu.cn)
Hainan Medical University China	Link- Gas Sensor Array Drift Dataset - UCI Machine Learning Repository
University California Irvine gas sensor datasets	Link- 1)OhioT1DM Dataset
OhioT1DM	2) Advanced Bolus Calculator for Diabetes (ABC4D) Research groups Imperial College London
ABC4D	Link- ECG signals (1000 fragments) - Mendeley Data
ARISES	Link-https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE81608.
ECG signal data on 218 patients	Link- National Health and Nutrition Examination Survey (kaggle.com)
RNA Sequencing Data	Link- OpASHA Other papers (worldbank.org)
National Health and Nutrition Dataset	Link- Diabetes Statistics - NIDDK (nih.gov)
Clinical data collection by OpASHA	Link- SREE DIABETES SPECIALITY CENTER
National Institute of Diabetes and Digestive and Kidney Diseases	
Sree Diabetes Diagnostic center Kurnool	Link-[PDF] First Experiences with the Identification of People at Risk for Diabetes in Argentina using Machine Learning Techniques Semantic Scholar
PPDBA program dataset	

used the glucose samples but also used the genes and image classification.

6. State of art deep learning and machine learning approaches in diabetes mellitus

Machine Learning and Deep Learning techniques always provide better results when applied to healthcare data worldwide. Both learning techniques are frequently used to solve real-life problems. These techniques are commonly used for diabetes detection and prediction procedures. Table 3 shows the review and limitations of ML and DL Classifiers in diabetes mellitus.

Finally, their commonly reviewed algorithm of ML and DL is used in this paper, as shown in Fig. 4. Here are several widely used algorithms:

Table 3
Critical Review of ML and DL Classifier in Diabetes Mellitus.

S. No.	References	Research Area	ML and DL Classifier	Performance
1	Ahmed et al. [1]	Machine Learning	SVM	94.8% Accuracy
2	Shokrehodaie et al. [2]	Machine Learning and Regression Neural Network	SVM RNN	99.78% Accuracy
3	Sharma et al. [3]	Extreme Machine Learning Techniques Amazon Cloud	ANN	90.57% Accuracy 72.23% Sensitivity 75.35% Specifity 98% Accuracy
4	Thesis et al. [4]	Sensor Technology and Machine Learning	LSTM	
5	Lekha et al. [5]	Machine Learning Data Mining	CNN PCA	95% Accuracy
6	Alkhodari et al. [6]	Machine Learning Neuropathy	SVM CNN	90.51% Accuracy of SVM 98.5% Accuracy of SVM 95% Accuracy
7	Al Jlaite et al. [7]	Machine Learning Deep Neural Network Image Processing	SVM PCA KNN	
8	Fazakis et al. [8]	Machine Learning Artificial Intelligence IoT	SVM	The AOC Curve value is 0.884%
9	Siddiqui et al. [9]	Glucose Monitoring Sensors	SVM ECG Signals	Pain Free monitoring of Diabetic patients for a day. 96% Accuracy
10	Thakkar et al. [10]	Fuzzy Logic Classification Random Forest Machine Learning	Fuzzy Neural Network Random Forest	
11	Fiarmi et al. [11]	Machine Learning Data Mining Clustering	Naive Bayes C4.5 Classification K- Means Clustering	68% Accuracy from the retinopathy model.
12	Suyanto et al. [12]	Machine Learning Dimensionality Reduction	KNN 5-Fold Cross Validation	98.07% Accuracy
13	Mamatha et al. [13]	Big Data Analytics Machine Learning	Gaussian Naive Bayes BIRCH Algorithm	The precision value from the model is 0.735. The recall value from model is 0.79.
14	Shuja et al. [14]	Deep Learning Machine Learning Oversampling Techniques	SVM MLP (Multilayer Perceptron)	94.70% Accuracy
15	Kazerouni et al. [15]	Machine Learning	KNN ANN SVM	95% Accuracy
16	Khanam et al. [16]	Machine Learning Regression	SVM Logistic Regression K-foldcross-	86% Accuracy with ANN

Table 3 (continued)

S. No.	References	Research Area	ML and DL Classifier	Performance
17	Chaki et al. [17]	Artificial Intelligence Machine Learning Diabetes Management Techniques	validation ANN Reinforcement Learning CNN	–
18	Ghosh et al. [18]	Machine Learning Cross Validation	Random Forest SVM Gradient Boosting	99.35% Accuracy using SVM
19	Tripathi et al. [19]	Machine Learning	KNN SVM Random Forest	87.66% Accuracy
20	Ye et al. [20]	Deep Learning Machine Learning Regression	SVM Gradient Boosting Logistic Regression	90.04% Accuracy from E-Nose System
21	Zhu et al. [21]	Machine Learning Gas Sensor Array Boosting	SVM XGB Boost Gradient Boosting Technique	90.55% Accuracy
22	Zhu et al. [22]	IoT Edge Computing Deep Learning	LSTM RNN	The IoT based model on glucose sensor data provide 77% accuracy
23	Gupta et al. [23]	Machine Learning Signal Processing	Decision Tree ECG Signals Classification	86.9% Accuracy
24	Li et al. [24]	Machine Learning RNA Sequencing	SVM Monte Carlo Feature Selection	94.2% Accuracy
25	Carlton et al. [25]	Quantitative surveying Qualitative interviews	Cognitive debriefing interviews Refining the descriptive system	This psychological study on diabetic patients increased people interest accuracy up to 80%.
26	Aguilera et al. [26]	Machine Learning Application Study	Regression Learning Bayesian Method Thompson Sampling Methods	In this learning methods this study taken on m-health app and more than 80% people interested in study.
27	Ayon et al. [27]	Deep Learning Cross Validation Methods	Deep Neural Network Ten-Fold Cross Validation Five Cross Validation	In this paper the accuracy is 98.35%. F1 Score Value is 98
28	Manizuraman et al. [28]	Machine Learning Regression	Logistic Regression Naive Bayes Decision Tree Random Forest	The Specificity value with tenfold cross validation is 98.80% 90.62% Accuracy achieved with Naive bayes. 90.62% Accuracy achieved with Random Forest

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Table 3 (continued)

S. No.	References	Research Area	ML and DL Classifier	Performance
29	Aada et al. [29].	Machine Learning	Naive Bayes Decision Tree KNN	Classifier is 94.5%. The Accuracy on the datasets is 94.8% The Precision value is 73.28% The recall value is 73.79%
30	Pranto et al. [30]	Machine Learning	Decision Tree KNN Random Forest Naive Bayes	The accuracy on Kurmitola Hospital dataset is 81.2% The accuracy on PIDD dataset is 77.9%
31	Gröschel et al. [31]	Tuberculosis Data Mining	Random Glucose Sampling techniques	Person diagnoses the diabetes with accuracy of 86%.
32	Nath et al. [32]	Artificial Pancreas System Glucose Monitoring System	Hovorka's Model Dolla Man's Model	–
33	Khan et al. [33]	Type-2 Diabetes prediction on 5-types of cell	Regulation of Insulin on the Five types of Cells	–
34	Vyas et al. [34].	Diabetes Risk Analysis	Swarm Algorithm PSO Algorithm	–
35	Yasen et al. [35]	Machine Learning Deep learning Data Analysis	ANN Dragon fly Algorithm Firefly Algorithm	The accuracy for diabetes is 79.77%. The accuracy for heart disease is 90%.
36	Thippa Reddy et al. [36]	Machine Learning Deep learning Soft Computing	Firefly Algorithm BAT Algorithm Optimization Techniques	The Accuracy get from firefly algorithm is 70%.
37	Olisah et al. [37]	Machine Learning Deep learning Data Handling	Random Forest SVM Deep Neural Network	97.3% Accuracy achieved with these algorithms.
38	Ganie et al. [38].	Machine Learning Cross Validation Methods	SVM Decision Tree SMOTE Logistic Regression	99.4% Accuracy achieved with these algorithms.
39	Yahayaoui et al. [39]	Machine Learning Deep Learning	SVM Random Forest	65.38% Accuracy achieved with SVM algorithm. 83.67% Accuracy achieved with Random Forest.
40	Dremin et al. [40]	Machine Learning Image Processing	Fuzzy Neural Network Random Forest	The polarization Index of Diabetic person is 95%
41	Wee et al. [52]	Deep Learning Machine Learning	CNN PCA LDA DNN	The Deep learning model provides the accuracy of 89%.
42	Al-Absi et al. [53]	Deep Learning IoT	Transfer learning	This model provides

Table 3 (continued)

S. No.	References	Research Area	ML and DL Classifier	Performance
43	Sarmun et al. [54]	Deep Learning	VGG-11 deep learning network, Weighted bounding box fusion (WBF), non-maximum suppression (NMS), Soft-NMS	impressive accuracy of 93%. This model provides the mean average precision (mAP) score of 86.4%
44	Khan et al. [55]	Deep Learning	Sequential Feature Selection (SFS) and Mutual Information (MI) feature selection	The random forest achieved an accuracy of 99.36%.

- 1) **SVM**-The SVM is used to classify the data and map it into high-dimensional feature space by separating the data with a hyper-plane. The hyperplane separates the data into different samples and classes. A decision boundary is created using kernel functions like linear, polynomial, and sigmoid. Mamatha et al. [13] focused on the kernel function is an essential aspect of SVM. It allows the value associated with the decision boundary to create the difference between various classes. The SVM model is so much simplified for diabetes detection. However, in this model, the possibility of over-fitting is also increased. This model can accurately classify the diabetic data based on test sets, training sets and validation sets without adding unnecessary complexity to any prevention and detection of diabetes.
- 2) **KNN**- KNN is an algorithm that categorises unlabelled data points depending on the similarity of various data points in a close prominent region. The KNN data points are labelled, and there are no issues in calculating the distance of data points. The problem arises when the KNN gets unlabelled data [7]. In KNN, two data points are taken, and a voting rule applies to the classification of datasets. Two standard methods are applied to categorize the unlabelled data points are-
 - a) Majority voting rule is used.
 - b) The Inverse distance weighting rule. The distance between the two points is also measured using the Euclidian distance formula.

The class with minimum distributed value gets the vote, and the vote is like an assigned label to the unlabelled data points.

- 1) **Decision Tree Classifier**-Maniruzzaman et al. [28] mentioned in the article decision tree has a tree structure that is the same as a tree. A tree has a structure of roots, nodes, branches, and leaves. The decision tree also has the same structure as decision nodes, leaf nodes, and branches. The leaf nodes and terminal nodes show the class label of a tree for a final prediction and split the dataset into decreasing levels of entropy. The tree started from a root node, the splitting of a node until it reached the end. The performance of the decision tree was measured into three points:
 - a) The maximum depth of a tree is shown as max_depth,
 - b) The minimum no of the split is shown as min_samples_split,
 - c) The maximum number of leaves in the tree is denoted as min_leaf_nodes.
- 2) **Principal Component Analysis**-Sharma et al. [3] more focused on the principal component analysis is an essential technique for dimensionality reduction. It takes the feature set and attribute set for dimensionality reduction. This reduction in dimensionality helps simplify the data and often leads to improved model performance

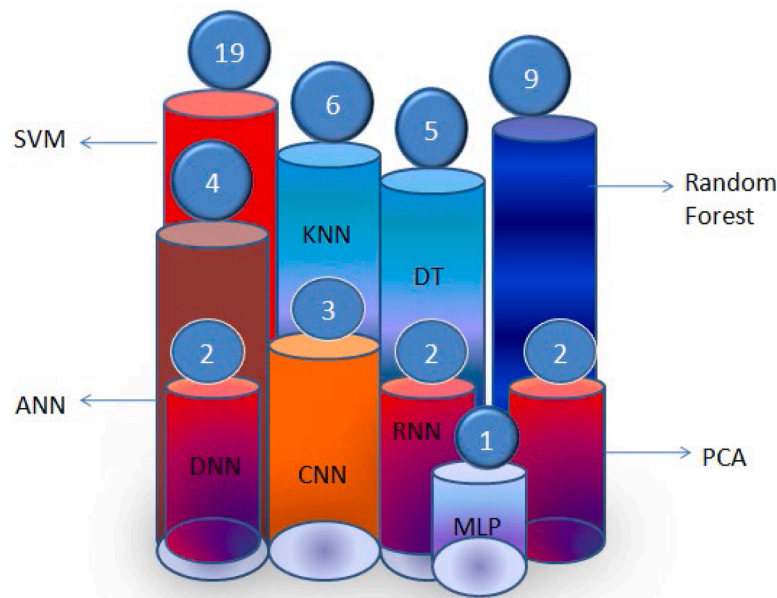


Fig. 4. Categorization of Algorithms uses Machine Learning and Deep Learning techniques analysed in this manuscript.

and computational efficiency. In the context of diabetes, PCA can be employed in several ways –

1. **Feature reduction**—Diabetes detection and prediction are based on features (e.g., glucose, insulin blood pressure, BMI, etc.). PCA can decrease the feature count by merging similar types of features into a smaller set of uncorrelated principal components. It reduces the multi-co linearity issues and supports model interpretability.
2. **Data Visualization**—In PCA, the high-dimensional diabetes data in 2D or 3D can be easily visualized by plotting the data points of the principal components.
3. **Pre-processing**—This is indulgently working on data pre-processing techniques; it's a step before feeding the data into machine learning models.

It performs the dimensionality reduction procedure to reduce the risk of overfitting.

- 1) **Deep Neural Network**- It relies on unsupervised learning that is used in the prediction and detection of many diseases and many health applications. This neural network is used for measuring blood sugar levels and early detection of diabetes, and it's essential for efficient management and prevention of complications in the context of diabetes. Deep Neural Networks are used in several ways:

1. **Data Collection**—This is the first step in DNN to assemble informative data on patients. The data set consists of blood glucose levels, age, Body Mass Index, Blood pressure, and other health-relevant indicators. This dataset is used to detect whether a person is diabetic or not.
2. **Network Architecture Model**—The DNN architecture model is very complex in its performance. It may consist of some other neural architecture, such as feed-forward neural networks, convolutional neural networks (CNNs) for image-based features, or recurrent neural networks (RNNs) for sequential data for the blood glucose detection of diabetes.
3. **Training of Model**- The diabetic dataset is categorized into a training set or validation set.

The DNN model is trained on the training set using an intensification algorithm like gradient descent to minimize the loss function of the diabetic dataset as mentioned in the article of Theis et al. [4].

- 1) **Random Forest**- Random Forest is a type of decision tree; this tree is also known as the form CART (Classification and Regression Tree). It is a collection of more decision trees. In this tree, each node sub-tree is connected to the other node of the tree. Alkhodari et al [6] has provide the concept of bagging and boosting is commonly used, and researchers are interested in using random forest and boosting techniques for diabetes prediction. Both Random Forest and Boosting are popular ensemble machine learning methods used for classification tasks like predicting diabetes. Whenever people apply a boosting technique in a random forest, it starts by building a baseline random forest model. The random forest worked on predictions depending on the features, such as the number of trees, maximum depth, and minimum samples per leaf.
- 2) **Multilayer Perceptron**-MLP works in the same way the human brain works. The human brain handles and responds to any situation the same way MLP does. MLP is the advanced version of ANN. A Multilayer Perceptron (MLP) is a form of artificial neural network that features several layers of nodes, also known as neurons, which are interconnected. It usually includes an input layer, several hidden layers, and an output layer. Within each layer, the neurons execute calculations on the incoming data. In a multilayer perceptron (MLP), neurons are organized into layers. Each neuron in these layers gets inputs from the previous layer, calculates a weighted sum of these inputs, adds a bias term, and then processes the data using an activation function. The architecture of an MLP is so dense it consists of more hidden layers and complex features from the data as given in the article Shuja et al. [14].
- 3) **Convolutional Neural Network**-CNN is a category of deep learning that works similarly to the feed-forward neural network. In CNN, the data is analyzed through transitional and rotational methods. This neural layer is applied to the convolutional operation input data. It also applies some filters to the input data. This network trained itself automatically based on features and patterns [6]. This algorithm is specially designed to process and analyze visual data, such as images and videos.
- 4) **Artificial Neural Networks**-ANNs work the same way humans respond to any situation. It processes the data based on computational techniques. The ANN applies the model to our problems and obtains high performance. The ANN algorithm maps the input data combined with appropriate output classes. According to Kazerouni et al. [15] the ANN consists of an Input layer, a Hidden Layer, and an

output layer. Its architecture indulges in a neural network with many artificial neurons, which is termed as each unit arranged in a sequence of layers.

- 5) **Recurrent Neural Network**-RNN worked for the sequential data. In RNN, the processing of sequential data depends on the time series. The time series depends upon from 1 to π . The RNN is called recurrent because it executes the same task for every element of the sequence. Zhu et al. [22] worked on-time computations, its output is dependent. Another process of thinking of RNN is that it captures the information that has been calculated so far. The RNN algorithm has four stages: the first one is the Input Stage, the second one is the hidden stage, the third is the weight, and the last is the output stage.

7. Discussion

In this manuscript, machine learning and deep learning techniques are reviewed. This Paper reviews the supervised, semi-supervised, reinforcement, unsupervised, and deep learning articles by analysing the results, techniques and methods. The main motive behind this Paper is to determine classification methods from a qualitative viewpoint that substantially facilitate future research efforts and increase awareness of the advancements in the domain of diabetes detection and prediction. In this review, the dataset of 55 articles is reviewed. The research of machine learning and deep learning will create automatic diabetes detection platforms.

This review encompasses a wide range of topics, including the exploration of various databases and their unique methods, classification techniques, and the diagnosis of diabetes. We also divide into the performance metrics of the models, which serve as indicators of model efficiency, accuracy, specificity, F1-score, and recall value. Additionally, we discuss the empirical-based research, which, despite its time-consuming nature, has proven to be the most effective in diabetes prediction and detection.

Ayon et al. [27] performed research on diabetes detection and prediction with the help of breath. The gas sensor is used for breath detection, and Acetone as a breath marker. A multivariate relevance vector machine trained the gas sensor on the database. In modeling, the temperature and humidity sensor gas sensor perform analysis by CGS-8 intelligent gas sensing and analysis systems. Gupta et al. [23] worked on diabetes detection using ECG signals. The ECG Signals experiment was performed using a total of 86 persons to test whether the person had diabetes or not. In total 86 persons, they found 35 persons have diabetes. The model contains ECG recording, Intrinsic time scale- decomposition (ITD), features, and classify the features of persons. Li et al. [24] performed research on human genes for the detection of Type-2 diabetes. Type-2 diabetes has 1600 single cells, and these 949 cells have Type-2 patients and 651 cells from everyday concepts. The selected features and algorithms used are KNN, SVM, and RF. After applying this algorithm, the results are provided optimized Type-2 diabetes-associated and performed optimized classification techniques.

Carlton et al. [25] has given three methods are used to perform the empirical research. Qualitative interviews will inform the draft PROM content. Cognitive debriefing on the interview data refines the draft from PROM content. A classification system has generated a smaller number of items from PROM. Aguilera et al. [26] performed the research studies on diabetic patients determine whether the diabetic person is in the depression or not. This study was performed on 276 adults. These adults are between 18 and 75 years old. They measure the depressive symptoms on a scale of 8. The depressive scale intervention was checked within six months, followed by a follow-up procedure. The PIDD dataset of diabetic patients was analysed using a deep neural network. The classification procedure in this research was performed using K-fold cross-validation [27]. Diabetes diagnostics employed the same PIDD database. The bootstrapping method was used to correct the dataset. The model comprised dataset selection, data pre-processing, feature extraction through PCA, and the application of the resample channel to train

the model with classifiers such as KNN, Naive Bayes, and decision trees. The model was prepared to forecast the prediction result [29].

The research based on female patients in Bangladesh has been used to practice diabetes detection. In this research, the two databases formed a comparative study with various machine learning algorithms PrantoB, et al. [30].has taken two datasets taken are PIDD, and second one is Kurmitola General Hospital Datasets. Eight samples for feature extraction used four machine Learning algorithms: KNN, Decision Tree, Random Forest and Naive Bayes the patients of Diabetes and tuberculosis who are residing in the urban slums of India. If a patient having tuberculosis is not getting proper treatment, then the person becomes a diabetic patient. The continuous supply of glucose makes the person diabetic. It will increase the risk factor if the person becomes diabetic patient. The random glucose sampling of a person with HbA1c features confirms the identification of undetected diabetes cases [31].

Khan et al. [33] their research focuses on insulin management, explicitly examining the regulation and modification of insulin across four to five cell types, named alpha, beta, delta, pp (f cells), and epsilon. These cells were analysed under various parameters, assessing their impact on insulin secretion, glucose regulation, and somatostatin secretion. Among these, the beta cell exhibited superior performance. Vyas Set al. [34], explored various diseases using datasets including digestive and kidney diseases, digestive disorders, and diabetes. So many multiple algorithms are used for diabetes detection. This paper used the ANN, Dragonfly algorithm and artificial bee algorithms. A new algorithm was implemented called and named it (ANN-DA). The ANN-DA algorithm provides better results as compared to the ANN algorithm. This algorithm predicts diseases like Hepatitis, Diabetes, breast cancer, blood donations and heart disease [35]. Thippa Reddy et al. [36] also focused on the classification of diabetes data. They have created a new algorithm for diabetic classification. This algorithm, named the firefly-BAT algorithm (FBAT), is a rule-based fuzzy logic-based prediction algorithm. The fuzzy logic-based system has been designed with the help of fuzzy-based rules. The process consists of a rule-based system of diabetes and decision and optimal rule generation via FF-BAT. Olisah et al. [37] performed research on diabetes detection using deep learning techniques. PIDD dataset used for detection. The prediction process is categorized into two sets, the first of which is the data filtration and cleaning process. In the second set, all predictive analyses were performed. In the first set, they take the PIDD database, remove the missing values, give feature importance and feature selection procedure, and provide the missing value imputation process. In the second set, data are tested and trained, and analysis of the basic model is performed to give the predicted results. The prediction of diabetes in a lavish lifestyle. In this paper, survey-based research was conducted. The sample data was collected from the population of Jammu & Kashmir by distributing the survey form to J&K. The data was collected from 1939 people. Rename this dataset into T1IDM Lifestyle datasets. Data processing techniques and data splitting procedures have been performed on it. Applying the K-fold cross-validation technique will result in the T1IDM [38]. Yahyaoui et al. [39] performed Diabetes detection using machine learning techniques involves taking a dataset and adding some original features. Bootstrapping methods are applied for data filtration, and a majority voting process is used to obtain results. Input data are processed using convolutional max pooling layers and feature mapping on dense layers, with Random Forest being used to obtain the results.

Khan et al. [45] worked on a groundbreaking multi-view clustering approach was introduced, leveraging concept factorization to enhance clustering performance. This novel method surpassed existing techniques, showcasing its efficiency in handling multi-view data. Khan et al. [46] developed Another innovative framework, DECCA, was developed based on Contractive Autoencoders. It demonstrated superior results in clustering high-dimensional document data, even on real-world datasets.

Debelee et al. [47] worked on the Commonly available datasets for skin lesion analysis were identified, exploring various machine-learning

techniques used for skin disease detection. The study examined the contributions and limitations of current state-of-the-art methods. Khan et al. [48] introduced a new multi-view clustering algorithm called MCNMF that utilizes manifold regularization to significantly improve clustering results as compared to other algorithms on various real-world datasets. The importance of accurate detection of diabetes is demonstrated by proposing a method using machine learning techniques for this purpose, testing the process on a diabetes dataset, achieving high performance compared to previous methods, and emphasizing the significance of early detection for effective treatments [49]. Rucci et al. [50] more focused challenges in detecting Type 2 Diabetes and Prediabetes present the development and evaluation of predictive models for identifying at-risk individuals in Argentina and emphasize the importance of early detection in individuals unaware of their condition. The global and local learning models of diabetes detection have been designed. Rufo et al. [51] developed a model by integrating diverse machine learning techniques like XGBoost and Naive Bayes for international learning. The local learning model consists of KNN, SVM and RBF for local learning. They apply this model to the Pima Indian Diabetes Dataset, which provides an accuracy of 99.5%. Overall, the paper offered a vast description of the diabetes dataset used in deep learning and machine learning techniques for the identification of diabetes. Table 4 shows the summary of related works, highlighting both contributions and limitations of this study.

8. Conclusion

Recognizing the pivotal role of ML techniques in advancement in the healthcare sector, particularly in disease prediction, the manuscript investigates the utilization of various ML and DL approaches for the early detection of Diabetes. The analysis offered in this work revolves around five key factors: the dataset used for Diabetes Mellitus, machine learning-based identification methods, performance measures, limitations in current diabetes detection research, and the overall status of diabetic research.

The study objective of this manuscript is to provide a detailed study on DM prediction, diabetes management procedures, and preventive techniques. This information is designed to serve as a practical resource for researchers in the field. The importance of a systematic approach and the inclusion of critical factors contribute to a more nuanced understanding of the subject matter. As the scientific community remains to engage in a comprehensive review, automated DM detection and self-management emerge as vital avenues for further exploration and development in diabetes research. The insights presented herein lay the foundation for ongoing works to address the multifaceted objection posed by Diabetes Mellitus on a global scale. Due to the rapid advancement of Artificial Intelligence and the Internet of Things, we would like to integrate further these advanced technologies for data processing for the early prediction of Diabetes Mellitus.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

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CRedit authorship contribution statement

Neha Katiyar: Writing – review & editing, Writing – original draft, Visualization, Methodology, Conceptualization. **Hardeo Kumar Thakur:** Supervision, Methodology. **Anindya Ghatak:** Supervision, Conceptualization.

Table 4
Summary of work with their contributions and limitations.

S. No	Authors Details	Contribution	Limitations
1	Ahmed et al. [1]	Their groundbreaking work is cantered on elevating the precision and dependability of diabetes prediction through the fusion of multiple machine-learning models. The authors innovatively crafted a unified machine learning framework that amalgamates the merits of diverse predictive algorithms, thereby enhancing the prediction outcomes for diabetes.	Fuzzy logic techniques are ingeniously employed to blend two to three ML algorithms.
2	Shokrehodaiei al. [2]	They conducted comprehensive experiments to validate the performance of their system. They provided a detailed analysis of the system's accuracy in glucose level detection, comparing it with traditional invasive methods and demonstrating its efficacy.	The VIS-NIS sensor size must be minimized so it turns out to be a wearable sensor.
3	Sharma et al. [3]	They proposed a novel service composition model within the cloud environment that integrates various healthcare services. This model allows for the efficient coordination of medical resources and services, optimizing the healthcare delivery process for diabetes patients and improving the overall efficiency and effectiveness of medical care provision.	. The sensitivity and relativity value and accuracy are less.
4	Thesis et al. [4]	They critically analyzed the integration of machine learning techniques with E-Nose technology, discussing how these combined approaches can increase the accuracy and reliability of non-invasive diabetes diagnosis. The review also explored future directions in machine learning algorithms for improving diagnostic performance and personalized healthcare solutions in diabetes management.	The approach they used for the longitudinal patient records is not applicable to the minimal time patients stayed in the hospital.
5	Lekha et al. [5]	They developed an advanced prediction model that combines process mining with deep learning techniques to enhance the in-hospital mortality prediction for diabetes patients in the intensive care unit (ICU).	. The MOSFET sensor for breath detection is not wearable. Use for clinical purposes is not possible.

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Table 4 (continued)

S. No	Authors Details	Contribution	Limitations
6	Alkhodari et al. [6]	They contributed by developing a machine learning-based method to screen for cardiovascular autonomic neuropathy (CAN) in diabetic patients, particularly those with microvascular complications. Their method utilizes 24-hour heart rate variability (HRV) data to diagnose complications of diabetes.	The approach required careful considerations on the clinical implementations.
7	Al Jlailety et al. [7]	They contributed to the practical relevance of machine learning techniques for analysing and converting data from inertial sensors worn by individuals. Their work not only improved the accuracy and efficiency of gesture detection but also integrated technologies in everyday health monitoring devices.	None
8	Fazakis et al. [8]	They created intelligent machine learning tools for the long-term risk of developing type 2 diabetes prediction. Their models can accurately assess an individual's risk, enabling early intervention and preventive healthcare measures.	The AOC curve value is so much less.
9	Siddiqui et al. [9]	They extensively reviewed non-invasive or painless blood glucose monitoring techniques from 2012 to 2016, examined the progress in the diabetes prediction area, and facilitated the connection between information technology and healthcare.	If there were fewer features, the accuracy rate could be higher. It takes only Five features.
10	Thakkar et al. [10]	The study incorporates understanding and using advanced analytical techniques to enhance the accuracy and reliability of diabetes prognosis. By dissecting and analysing these techniques, more sophisticated and effective tools for diabetes management and research are developed.	Precision and recall value is not calculated. Feature description is absent in the paper.
11	Fiarmi et al. [11]	They contributed by comprehensively analysing diabetes data to interpret the patterns and correlations related to diabetes complications. They applied data mining techniques and the effective use of clinical data to improve the prognosis and treatment of diabetes diseases.	The model used in the paper provide less accuracy.
12	Suyanto et al. [12]	They use medical datasets to predict diabetes. This work provides significant	In this paper, splinting the dataset into clusters is impossible.

Table 4 (continued)

S. No	Authors Details	Contribution	Limitations
13	Mamatha et al. [13]	results over traditional methods, making it a more reliable and effective tool for early diabetes diagnosis and prevention. The authors worked on big data and applied data mining in healthcare, specifically focusing on the detection of diabetes. They contributed to the potential of big data analytics in enhancing the understanding, analysis, and prediction of diabetes; they were more focused on personalized healthcare solutions.	It shows that 768 samples are big data. These sample values are not considered as big data.
14	Shuja et al. [14]	They implemented the Synthetic Minority Over-sampling Technique (SMOTE) to address data imbalance, a common challenge in medical data analysis. This technique significantly improved the predictive performance of their model with the data mining process.	The model they provide for DL and ML techniques needs to be clearly defined.
15	Kazerouni et al. [15]	They explored the potential of long non-coding RNA (lncRNA) expression for predicting Type 2 diabetes mellitus (T2DM) and detected diabetes on an RNA molecular basis.	The model used in the paper needs to be more clearly defined.
16	Khanam et al. [16]	They contributed to performing detailed comparative analyses on different machine learning algorithms to assess their efficiency and accuracy in predicting diabetes. They provide valuable insight for better prediction and management of diabetes.	The accuracy is less than 77% on the train and test split phase.
17	Chaki et al. [17]	They performed a systematic review to evaluate the effectiveness of machine learning (ML) and artificial intelligence (AI) techniques in the detection and self-management of diabetes mellitus and highlighted the research gaps in this field.	It takes less than no paper to do a diabetic review.
18	Ghosh et al. [18]	They contributed by performing research on diabetes management tools.	The Data Pre-Processing techniques are used for better results.
19	Tripathi et al. [19]	They conducted an in-depth analysis of various machine learning algorithms to determine the most effective approach for predicting early diabetes. Their research contributes to the optimization algorithms for healthcare	The Accuracy must be improved.

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Table 4 (continued)

S. No	Authors Details	Contribution	Limitations
20	Ye et al. [20]	applications, improving the predictive accuracy for early intervention in diabetes.. They contributed by designing the model of an e-nose system that could achieve high levels of accuracy and precision in predicting blood glucose levels. This advancement not only proves the system's effectiveness in real-world applications but also paves the way for further research and development in non-invasive glucose monitoring technologies.	The experiment was performed on fewer people. It takes only 41 people. More people take the time to get better results.
21	Zhu et al. [21]	Their work showcases the effective use of a gas sensor array in identifying the breath markers of diabetic patients. This approach provides a non-invasive, real-time method for diabetes detection, contributing to easier and more patient-friendly diabetes monitoring and management.	The working with ethanol gasses. Its clinical implementation required some precautions.
22	Zhu et al. [22]	They innovatively combined the Internet of Medical Things (IoMT) with deep learning algorithms to develop a real-time blood glucose prediction system. This system represents a significant advancement in the remote and continuous monitoring of glucose levels, enhancing diabetes management through timely and accurate data analysis.	The accuracy rate is less. Few methods and techniques apply to the datasets.
23	Gupta et al. [23]	Their research contributes to remodeling the accuracy and reliability of diabetes screening. By leveraging ECG signals commonly collected in clinical settings, the framework enhances the capability to identify at-risk individuals, thus supporting timely and appropriate medical intervention.	The drawbacks of this paper accuracy must be improved.
24	Li et al. [24]	They contributed mixed single-cell sequencing data to identify biomarkers for type 2 diabetes. This innovative approach allows for a more detailed and nuanced understanding of the cellular and molecular mechanisms involved in the disease, leading to the discovery of specific biomarkers that can be used for diagnosis,	The number of cells in a group needs to be balanced. It looks like a raw data set.

Table 4 (continued)

S. No	Authors Details	Contribution	Limitations
25	Carton et al. [25]	treatment, and monitoring of type 2 diabetes. They Contributed by elaborating a new preference-based model to quantify the significance of hypoglycaemia on the quality of life of people with diabetes. This addresses a significant gap in diabetic care and measures how hypoglycaemia affects patients' lives.	The Full psychometric Testing is required for this model.
26	Aguilera et al. [26]	They created a detailed clinical trial protocol to test the DIAMANTE app's effectiveness rigorously. This contribution is vital for providing empirical evidence on how machine learning and mHealth solutions improve health outcomes, specifically by increasing physical workload in patients with diabetes and depression.	It takes a lot of time for this research, approxsix months.
27	Ayon et al [27]	The comparison performance of their deep learning model with traditional machine learning algorithms provides beneficial insights into the advantages and disadvantages of deep learning in the context of medical prediction. This comparative study helps to highlight the features of deep learning in enhancing the predictive analytics models for diabetes.	The present class of dataset is 259, which is less.
28	Manizurraman et al. [28]	The study evaluated various machine learning techniques and their performance in diabetes detection. By assessing these models on various metrics, the most effective for diabetes classification and prediction was determined, thereby guiding future research and clinical practice in this area.	The model is vast because it takes so many algorithms.
29	Aada et al. [29]	They provided a comprehensive analysis and comparison of the performance of these machine learning techniques in predicting diabetes.	The Accuracy get from using decision tree is so much less.
30	Pranto et al. [30]	They contributed a thorough comparative analysis of various machine learning algorithms that accurately predicted diabetes. This analysis not only highlights the most suitable predictive	The ANN algorithm is used for this extension of the research work.

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Table 4 (continued)

S. No	Authors Details	Contribution	Limitations
		models for diabetes prediction but also contributes to the broader field of medical informatics. Their research demonstrates that regional and demographic factors can influence the selection and performance of machine learning methods in healthcare applications.	
31	Grocheal et al. [31]	They investigated the efficacy of random glucose sampling for screening diabetes in disadvantaged populations, specifically among tuberculosis patients in urban slums in India. This approach addresses the need for accessible and practical diabetes screening methods with limited resources.	Accuracy must be improved.
32	Nath et al. [32]	They contributed a detailed review of different physiological models designed for understanding and simulating Type 1 Diabetes Mellitus. This contribution is accurate as it consolidates knowledge on modeling approaches that can replicate the physiological processes of T1DM, aiding in the research and development of treatment methods.	The noise cancelation process is required for diabetic people.
33	Khan et al. [33]	The article also explores the therapeutic potential of these islet peptides in treating type 2 diabetes. Because these peptides can be targeted or mimicked in therapeutic interventions, they contributed to the vast field of diabetes research, which is developing novel treatment strategies that could enhance beta cell function and improve glycaemic control in type 2 diabetes patients.	This paper requires a model for cell separation in different types, such as Alpha cell, Beta Cell, and Delta Cell.
34	Vyas et al. [34]	The article critically evaluates the effectiveness, strengths, and limitations of different predictive analysis techniques in the context of diabetes prospect assessment. This evaluation is crucial for understanding how these methods can be optimally applied and further developed to enhance the accuracy and reliability of diabetes risk predictions,	They take less paper for review.

Table 4 (continued)

S. No	Authors Details	Contribution	Limitations
		ultimately contributing to better preventive healthcare strategies.	
35	Yasen et al. [35]	They created an optimized neural network model applied to medical prediction tasks, demonstrating its potential in accurately diagnosing or predicting medical conditions. This contribution is significant as it provides a practical example of how advanced optimization algorithms are used in neural networks in the healthcare domain.	The optimization techniques may be used along with this algorithm.
36	Thippa Reddy et al. [36]	The use of the Firefly Algorithm to optimize the rule base of the fuzzy logic classifier represents an innovative approach in the field. This method not only improves the classifier's performance but also contributes to a broader understanding of how bio-inspired algorithms can be effectively applied in optimizing complex systems, particularly in the healthcare domain.	The accuracy could be better.
37	Olisah et al. [37]	They highlighted the critical role of data pre-processing in the machine learning pipeline for diabetes prediction and diagnosis. They explored various pre-processing techniques to enhance the quality and reliability of the data before it is used in machine learning models, showing how proper data preparation can significantly impact the accuracy of the predictive outcomes.	They take only adult persons' data for this implementation. Data on children and older people must be added.
38	Ganie et al. [38]	Their research emphasizes the significance of lifestyle indicators in predicting type-II diabetes mellitus, highlighting how factors such as diet, physical activity, and body weight are critical in developing and managing the disease. The study contributes to understanding how lifestyle data can be effectively utilized in predictive models to aid in early diagnosis and preventive healthcare strategies.	The Voting techniques they used can't provide accurate or valid results.
39	Yahayaoui et al. [39]	They contributed to enhancing predictive analytics in the healthcare sector by demonstrating how advanced data processing	The no of diabetic person data is so much less.

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Table 4 (continued)

S. No	Authors Details	Contribution	Limitations
40	Dremin et al. [40]	and analysis techniques can be employed to predict diabetes more effectively. Using sophisticated algorithms in their DSS represents a significant step forward in developing intelligent health monitoring and management tools. This contribution is significant in showcasing how machine learning can enhance the analysis and interpretation of complex imaging data, leading to more accurate and efficient diagnosis of skin-related diabetes complications.	The data taken by less no of patients.
41	Rufo et al. [44]	Develop an optimal and accurate diabetes diagnosis model based on machine learning algorithms.	Lack of enough data on recommended indicators like OGTT and HbA1c due to their expensiveness. The inclusion of invasive diabetes indicators may limit the application of self-testing. They limited generalization capability due to only having Ethiopia as a proxy for ethnicity.
42	Khan et al. [45]	It introduced a unique multi-view clustering technique to address the challenges in clustering multi-view data. That created a concept factorization method for consensus manifold regularization models to effectively factorize the original data matrix and value the shared geometrical structure from multiple-view data.	The suggested method leverages the manifold structure to preserve a uniform geometrical representation across various data spaces effectively.
43	Khan et al. [46]	They propose advanced techniques to avoid uninterested clusters by computing an accurate term in the loss function that penalizes the cluster.	Future studies ought to more meticulously evaluate the impact of recurrent neural networks on document clustering. Integrating recurrent neural networks with CtAE could lead to enhanced clustering accuracy. No identified financial conflicts or personal relationships might have affected the findings presented in this paper.
44	Debelee et al. [47]	Summarize the open challenges in skin disease and cancer detection and classification.	Lack of detailed comparison with other methods. Sole reliance on a single dataset. The absence of information on computational resources is required.
45	Khan et al. [48]	They contributed a new multi-view clustering method that depends upon NMF and manifold regularization, introduced a new function that integrates	Non-convex optimization problem. Difficulty in calculating the global minimum due to the non-convex nature of the objective function. Handling missing data by

Table 4 (continued)

S. No	Authors Details	Contribution	Limitations
46	Haq et al. [49]	NMF and manifold regularization and focused on multi-view data's complementary behavior to conserve the data space's local geometrical shape. They include developing a diagnosis system for accurate diabetes detection in the e-healthcare environment, enhancing prediction accuracy, proposing new approaches for effective diabetes detection, and evaluating classification performance based on different feature sets and cross-validation methods.	filling it with average feature values. The study's limitations are not explicitly stated in the paper. The paper focuses on the proposed method for detecting diabetes using ML techniques and highlights its achievements and potential future applications.
47	Rucci et al. [50]	The study objectives are to develop and assess predictive models to identify people at risk for Type 2 Diabetes (T2D) and Prediabetes (PD), specifically in Argentina, present the background for the work, describe the database processing, analyses the results obtained, and present conclusions and possible future work.	Requirements of large datasets to enhance the quality and representativeness.
48	Wee et al. [52]	The study objectives are to investigate and discuss the usability of ML and DL approaches in diabetes identification/ classifications. They highlight the future in the research of feature selection techniques densely used to improve the results of diabetes identifications.	Flaws and disadvantages in existing datasets such as inaccurate data or few recorded samples. Fluctuation in values of different recorded classes affecting machine learning models. Lack of a standard procedure for collecting non-invasive measurements for diabetes classification.
49	Al-Absi et al. [53]	The study objectives include developing an advanced predictive model for diabetes diagnosis using retinal images, addressing the limitations of current diagnostic methods, utilizing a massive dataset from Qatar Biobank and Hamad Medical Corporation to enhance accuracy, and achieving high precision, sensitivity, and specificity in distinguishing diabetic patients from the control group.	Absence of comparison with human-level intelligence in diabetes screening. Lack of public availability of datasets.
50	Sarmun et al. [54]	Accurate and early detection of Diabetes on Foot Ulcers through image analysis can prevent amputations and fatalities. The Weighted Bounding Box Fusion (WBF) approach, combining YOLOv8m and FRCNN-	Lack of diversity in non-DFU terms in the dataset. Predominance of white individuals in the data.

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Table 4 (continued)

S. No	Authors Details	Contribution	Limitations
51	Khan et al. [55]	ResNet101, significantly outperformed previous benchmarks, achieving a MAP score of 86.4% on the DFUC2020 dataset. The study achieved remarkable accuracy rates of 99.35% with ANN on the PIMA dataset and 99.36% with RF on the early diabetes risk dataset. The selected features using Sequential Feature Selection (SFS) significantly improved the prediction performance of machine learning models.	The machine learning models is trained and varies due to input attributes and their relationship with the target class label. The results of some methods decreased due to the reduction in the number of features.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Neha Katiyar, Research Scholar, School- School of Computer Science Engineering and Technology, Bennett University, India, Email- nehakatiyar20194@gmail.com, Research Area (s), Machine Learning, IoT, 6 G



Dr. Hardeo Kumar Thakur, Associate Professor, School- School of Computer Science Engineering and Technology, Bennett University, India, Email-Hardeo.Thakur@bennett.edu.in, Research Area(s), Data mining, Dynamic Graph Mining, Data Analytics



Dr. Anindya Ghatak, Assistant Professor, School- School of Computer Science Engineering and Technology, Bennett University, India, Email- anindya.ghatak@bennett.edu.in, Research Area(s), Functional Analysis, Operator Theory and Operator Algebras and its application in Quantum information theory.